

Evidence for cyclicity between anaphors and intensifiers

Noa Bassel (Hebrew University of Jerusalem)

Diachronic studies of the English *x-self* anaphor observe that its source is an early *self*-intensifier, compounded with a pronoun. Recent work showed that the Arabic anaphor *nafs-x* acquires an intensifying meaning when the anaphoric convention is relatively based, suggesting a reversed process. I propose that both processes can be seen as stages in a cyclic change where the pronoun + *self* compounding is a stage of renewal. The cycle explains a range of related phenomena across languages.

1. Introduction

Anaphors that express coreference in a local setting, such as English *x-self*, have raised interest for showing cross-linguistic regularity and diversity at the same time. On the one hand, they map to recurring morphological patterns and seem to follow a rule-based system in their distribution. On the other hand, they are intertwined in form and function with a range of elements of different linguistic categories, including *self*-intensifiers (e.g., *the queen herself*), focus particles, and middle voice morphology (Faltz 1977; Kemmer 1993; Déchaine and Wiltschko 2017; König and Siemund 2000a).

Faltz (1977) noted that it is a non-trivial task for speakers to draw a line between what linguists see as the core use of anaphors and other expressions of a similar form, even in light of completely different distribution and meaning. While each element in this (non-exhaustive) list has been the center of extensive literature, it has remained an open question why anaphors exhibit this network of formal and semantic relations across many different languages.

In the particular case of anaphors and intensifiers, there is a well-documented diachronic link in the history of Germanic and Romance languages, in which intensifiers constitute building blocks for anaphoric expressions (Keenan 1994; König and Siemund 1996, among many others). However, this explanation does not naturally extend to anaphors that emerge directly from lexical sources and acquire an intensifier use later, as in certain Semitic languages (Bassel 2024). One question that rises in this respect is whether the shift from intensifiers to anaphors is unidirectional, on a par with the theory of grammaticalization (Hopper and Traugott 2003), or a potentially unlimited process of reanalysis (Langacker 1977; Eckardt 2006) with no binding point of departure. The answer to this question should project more broadly on the source of the homophony between anaphors and intensifiers across languages, as well as the links to other elements.

The current paper proposes that anaphors and intensifiers are part of a cyclic historical change guided by principles of economy. The outline of this change is as follows: anaphors lose their referential capacity by reduction processes, leading to intensifiers and focus particles on one route, and to clitics and voice morphemes on the other. As these processes advance, languages may restore the anaphoric function by combining a referential pronoun with an older intensifier, which sets a new cycle in motion. The diverse typology of related elements noted by Faltz (1977) and others illustrates different stages in this process.

In what follows, Section 2 introduces the theoretical notion of cyclic change and presents the current proposal of a reflexive-intensive cycle. Section 3 lays out data for three historical stages in more detail and shows how they connect under a cyclic analysis. Section 4 concludes the paper.

2. Anaphors and intensifiers

A particular puzzle in the research of anaphoric expressions concerns the link between *self*-anaphors and *self*-intensifiers, illustrated below with English examples.

- (1) The queen invited ***(herself)** to the party. (anaphor)
- (2) The queen (**herself**) came to the party. (intensifier)

The *self* expression in (1) demonstrates a standard use of a locally-bound anaphor occurring as an argument of the verb *invite*. In (2), the same expression is adjacent to the noun *the queen* and its contribution is mostly pragmatic. More specifically, it generates an inference that other people of a lower profile than the queen also came, which is reminiscent of the additive effect known from the focus particle *even* (Crnič 2011). The link between the two *self* forms seems random on the surface, yet it exists in many languages beyond those related to English. König et al. (2013) found formal identity between anaphors and intensifiers in 94 out of 168 languages (56%) in a representative sample from different families and continents. The following sentences present the same homophony for the Modern Standard Arabic anaphor *nafs-x* (3a), which is also used as an intensifier (3b).

- (3) Modern Standard Arabic (Semitic, Middle East and Northern Africa)

- (a) *Anaphor*:

at-tarix jukarriru nafs-a-hu
the-history repeats.3SG.M self-ACC-3SG.M
'History repeats itself.'

‘The plans do not repeat themselves, the wars do not repeat themselves.’

(b) *Intensifier*:

w-al kitab-u al-muqaddas-i **nafs-u-hu** ðukir-a
 and-the book-NOM the-holy-GEN self-NOM-3SG.M mentioned.3SG.m-PERF
 fi-hi ma-la jaqel ʕan ʕifrina muðʕaharaten.
 in-3SG.M NEG-NEG less from twenty manifestation.PL
 ‘and the Bible itself mentioned no less than twenty demonstrations.’

(ArTenTen12)

The history of English offers a simple explanation for this trend in a documented diachronic link between anaphors and intensifiers (Penning 1875; Farr 1905; Visser 1966; Mitchell 1979; Keenan 1994; König and Siemund 1996; 2000b; van Gelderen 1996; 2000; Peitsara 1997; Lange 2001; Bergeton and Pancheva 2012). Historical corpora reveal that *self* morphemes existed in Old English only as intensifiers, and that their anaphoric use emerged more recently due to compounding with a pronoun, as seen in (4).

(4) Old English (Germanic, 5th-10th centuries CE)

Judas **hine** **selfne** aheng.

Judas him.ACC self.ACC hang

‘Judas hung himself.’

(Visser 1966 p.423)

This diachronic path is motivated by synchronic analyses that describe the meaning of both anaphors and intensifiers in terms of identity (Moravcsik 1972; Eckardt 2001; Hole 2006; König and Gast 2006; Reuland and Winter 2009; Charnavel and Sportiche 2022; Bassel 2024). It was previously assumed that the reflexive > intensive path reflects the only available order of emergence (e.g., König and Siemund 2000b). However, historical data involving the Arabic anaphor *nafs-x* and cognates in Hebrew and Aramaic suggest an opposite order (Bassel in preparation; Bassel and Keshev 2025).

Combining the information from English and Arabic seems to indicate that anaphors and intensifiers emerge through two possible directions of semantic change, in which the anaphor evolves from the intensifier and vice versa, as in Figure 1.

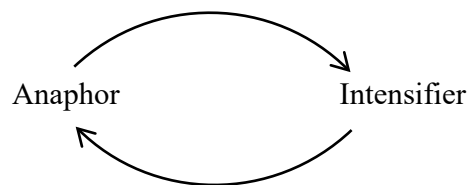


Figure 1: A multidirectional analysis of anaphors and intensifiers

This result is surprising from the standpoint of grammaticalization theory, which speaks of unidirectionality in language change (Roberts 1992; Hopper and Traugott 2003). However, works that center on the notion of reanalysis describe changes that are not predetermined in terms of directionality (Langacker 1977; Campbell 2000; Lightfoot 2006; Eckardt 2006). These are processes in which grammar generates more than one possible output per structure, where the choice between them is either arbitrary, filtered by the conversational context (Heine 2002), or motivated by grammatical principles, like considerations of economy (Lightfoot 1979; van Gelderen 2004; Kiparsky 2012). In this line of analysis, semantic change is unidirectional if it is motivated and unlimited otherwise (Bar-Asher Siegal, forthcoming).

Is the interaction between the anaphor and the intensifier motivated, and if so, in which direction? In the current paper, I propose to consider these changes as part of a cyclic process, in which the seemingly reversed clines are two unidirectional phases motivated by economy, pictured in Figure 2. Arabic demonstrates the formation of an anaphor from a lexical phrase, which then spreads to the meaning of the intensifier. The English process is a stage of renewal that forms a new anaphor.

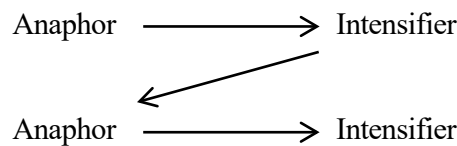


Figure 2: A cyclic analysis of anaphors and intensifiers. The horizontal lines represent reduction of the pronominal component, and the diagonal represents compounding with a new one

This analysis has three main advantages. First, it offers one explanation for the anaphor-intensifier homophony in English and Arabic. Second, it explains the asymmetry in which anaphors can extend their meaning to that of intensifiers with no formal change, while intensifiers merge with a pronominal element to acquire an anaphoric status. Finally, it sets anaphors and intensifiers within a broad range of well-known cyclic phenomena such as negation and tense marking. Such processes were previously explained by rules that guide speakers to assign simplified representations to complex input for economy considerations (e.g., Lightfoot 1979; van Gelderen 2004; 2011). A cyclic analysis of anaphors and intensifiers provides further support for this general insight and has a capacity to explain related phenomena as additional stages in the process.

2.1 Linguistic cycles

In the historical sense, a cyclic process is one where a semantic construction undergoes successive steps of reduction that conclude in a renewal of the original semantic function with

new linguistic material. Well-known examples include periphrastic markers of tense and aspect (Tauli 1956; Bybee et al. 1994; Krug 2011; van Gelderen 2011), modal verbs (Gergel 2009, Marušič & Žaucer 2016, Chatzopoulou 2019), negation (Jespersen 1917; Givón 1978; Zeijlstra 2004; Dahl 1979; Croft 1995; Bar-Asher Siegal and De Clercq 2019; Bar-Asher Siegal 2020; Hansen 2011; Breitbarth et al. 2020), and shifts from progressive to imperfective (Deo 2015), definiteness to person marking (Lyons 1999; van Gelderen 2011), pronoun to agreement (Greenberg 1978; Fuß 2005; van Gelderen 2011; Maddox 2021), pronoun to copula (Katz 1996). A few abstract illustrations for documented linguistic cycles are given in Table 1.

Phenomenon	Cycle
Subject agreement	noun > pronoun > agreement > $\emptyset$
Object agreement	pronoun > agreement > $\emptyset$
Determiners	demonstrative > definite article > “Case” > $\emptyset$
Negation	negative argument > negative adverb > negative particle > $\emptyset$

Table 1: Documented cycles (van Gelderen 2011: p.8)

Van Gelderen (2011) explains the recurrence of such changes as a general consequence of syntactic computation under cognitive constraints known collectively as economy principles. The argument builds on the axiom that any operation in phrase structure raises the complexity of its input. The basic operations of minimalist syntax, merge and move, present an obvious example: the merge of two objects A+B is a construction made of three objects (A, B, AB), while movement duplicates the features of one object to a second position. Historical changes like the ones in Table 1 balance the computational load through stages of simplification that reduce the number of objects. Economy Principles are rules of grammar that capture reduction under this general motivation.

An early formalization is the Transparency Principle from Lightfoot (1979), quoted in (5), which limits the possible delta between a string’s phonetic realization and its underlying structure.

(5) *Transparency Principle (Lightfoot 1979: p. 344):*

Derivations may be of a limited degree of complexity.

A transparency principle predicts that representations with many covert operations will be disfavored compared to ones that are more straightforward. Roberts and Roussou (2003) and

van Gelderen (2011) proposed more specific rules in this fashion, which speak of a motivation to interpret a phrase as a head (6), and a target of movement as a base position (7).

(6) *Head Preference Principle* (van Gelderen 2011: 15) :

Be a head, rather than a phrase.

(7) *Late Merge Principle* (van Gelderen 2011: 17; following Roberts and Roussou 2003):

Merge as late as possible.

According to van Gelderen, these principles govern historical processes and reinforce the general line of progression in (8).

(8) *Reduction cycle* (van Gelderen 2011: p.6)

Phrase > head > clitic > affix > Ø

If reduction is motivated by economy, it is expected to be unidirectional, which means reduced semantic functions could only be restored if they are encoded by new morphemes. Renewal is in itself motivated if it provides a simplified structure for a semantic function that is expressed by lexical means.

2.2 A reflexive-intensive cycle?

The main claim of the current paper is that the formation of anaphors and intensifiers corresponds to motivated steps of reduction and renewal. I base this claim on three lines of evidence, which I will outline here briefly to anticipate a more detailed presentation in the following section.

I will first present evidence for both directionalities in the anaphor-intensifier shift and show that both conform to economy: intensifier-to-anaphor processes turn a phrase into a head, while anaphor-to-intensifier processes reduce a pronominal suffix to agreement marking. Then I will argue that expanding the historical scope of investigation beyond these processes reveals a cyclic application of compounding and reduction.

For example, the history of French, like that of English, includes compounding between a pronoun and an intensifier *même* (Waltereit 2012). Like other Romance intensifiers, *même* is a derivative of the Latin intensifier *ipse*, which had earlier pronominal phases (König and Siemund 1999; Weiss 2009; Carlier and De Mulder 2010). In the other direction, *x-même* replaced the monomorphemic variant as the common intensifier. Taken together, these changes demonstrate two cycles, seen in Figure 3.

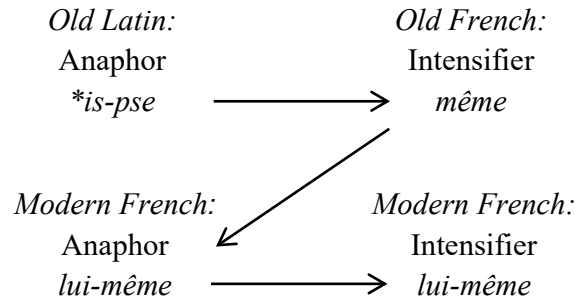


Figure 3: A reflexive-intensive cycle in French

We learn further from Arabic and German that intensifiers may reduce even further into modifiers meaning ‘same’ or ‘even’, which are respective developments of the intensifiers *naḡs-x* and *selbst* (Eckardt 2001; Bassel in preparation; Kayam 2024). Both of these meanings are also attested for *même*, having arrived as a bundle with the intensifier from Latin. Based on the chronology in Arabic and German, we may conclude that focus particles are a continuation of the shift from anaphor to intensifier, which suggests the progression in (9).

(9) Anaphor > intensifier > focus particle, ‘same’

The third line of evidence arrives from a parallel process that reduces anaphors to voice markers and possibly to nothing. This process, illustrated in (10), has already received a cyclic analysis in the literature (Kemmer 1993; Maddox 2021), but was so far discussed in separation from the anaphor-intensifier shift.

(10) Anaphor > clitic > middle voice > ∅

The most obvious relation between these processes is that anaphors that reduce to voice markers through the process in (10) often regenerate based on intensifiers and focus particles, the outcomes of the process in (9). Combining the two together forms the cycle in Figure 4.

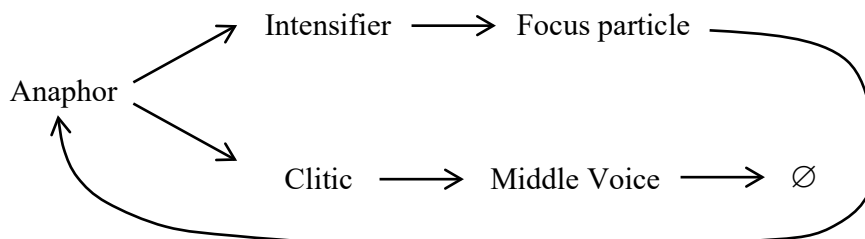


Figure 4: A reflexive-intensive cycle

To reiterate, the proposed cycle begins in the compounding of a pronoun with an additional morpheme, creating a complex anaphor. The anaphor has two possible paths of

disjoint one. In this capacity, *self* became a common addition to coreferential pronouns, with a rise of 750% in corpora frequency between the years 1250 and 1700 CE. (Keenan 1994; Peitsara 1997; Lange 2001). Throughout this period, *self* turned from an inflecting adjective into a fixed nominal head and gradually became more dependent on the pronoun. Instances of pronoun-*self* combinations before and in the course of this change are seen in (13-14).

(13) Old English:

ac wundorlice swyðe geeadmedde Crist **hine sylfne**.
 but wondrously much humiliated Christ him.ACC self.3SG.ACC
 ‘But Christ humiliated himself greatly.’ (Faltz 1977: 54)

(14) Middle English:

Hys ryȝt hand vp he lyfte and blessed **hym-self** stedfastly.
 his right hand up he lifted and blessed him-self steadfastly
 ‘He lifted his right hand up and blessed himself steadfastly.’ (Keenan 2002)

Interpreting such strings as compounds is economical since it turns them from phrases to a heads, as illustrated in (15).

(15) *Intensifier to anaphor*:

[DP [D him] [AP self]] > [DP himself]

This process was previously taken to explain the homophony between anaphors and intensifiers across languages (Kemmer 1993; Heine 1999; König and Siemund 2000b), alongside the possibility of a parallel emergence (Anghelescu 2011). However, in Bassel (2024; in preparation) I argued that a reversed path of change is also attested.

3.2 Anaphors to intensifiers

3.2.1 Semitic languages

Semitic anaphors belong to a class of anaphors that contain body-part expressions, known as ‘head reflexives’ due to Faltz (1977). A few examples are listed below, all have a parallel use as intensifiers.

Modern Hebrew	Palestinian Arabic	Standard Arabic	Amharic (Leslau 1995)
<i>acm-i</i> ‘my bone’	<i>ħal-I</i> ‘my state’	<i>naḥs-i</i> ‘my soul’	<i>ras-e</i> ‘my head’

Table 1: Body-part reflexives/intensifiers in Semitic languages

Given what is known from English, we may assume that these expressions start out as intensifiers and proceed to become anaphors. However, corpus data that probes into the use of *nafs-x* anaphors before they became a primary reflexive strategy does not support this directionality. In Bassel (in preparation) I focused on two corpora: the Hebrew Bible and the Qur'an. The languages and further details are listed below.

(16) *Data sources:*

- (a) Biblical Hebrew, 1st millennium BCE, 306,757 words
- (b) Qur'anic Arabic, 7th century CE, 77,797 words

The Qur'an shows the use of *nafs-x* in one of the earliest sources of Classical Arabic. The Bible features an even earlier use of the cognate *nap̄š-x*. If the directionality in which *nafs-x* emerged matches the one observed for *x-self*, we would expect to find intensive uses of the *nafs* morpheme in its basic form prior to the anaphoric use. Alternatively, we would find the compounded form in intensive contexts at least as early as it occurs in anaphoric contexts. Surprisingly, none of these options is realized in the data. A search of the respective lexemes *neṗēš* (נֶפֶשׁ) and *nafs* (نفس) in the Bible and the Qur'an retrieved no intensive uses, although *nafs* does have a common 'same' use in contemporary Arabic.

The contexts in which the basic lexemes occur in both sources suggest that the readings available for them in the relevant periods were their lexical meanings, mainly 'soul' and 'life', alongside an indefinite pronoun reading, which emerges under determiners (e.g., *kol neṗēš* 'everyone'). When searched with pronominal suffixes, Hebrew *nap̄š-x* turned up 99 anaphoric instances (0.32/1000w) and zero intensive ones, while Arabic *nafs-x* had 164 anaphoric instances (2.11/1000w) and 16 intensive ones (0.2/1000w). Table 2 summarizes the tokens and frequencies for each form.

		Anaphoric use		Intensive use	
		Instances	Frequency	Instance	Frequency
<i>neṗēš</i> (נֶפֶשׁ)	Base	0		0	
	Suffixed	99	0.32	0	
<i>nafs</i> (نفس)	Base	0		0	
	Suffixed	164	2.11	16	0.20

Table 2: Anaphoric and intensive occurrences of *neṗēš/nafs* and frequency for 1000 words

The significant gap seen in Arabic between anaphoric and intensive uses of *nafs-x* is also found in the Jewish Babylonian dialect of Aramaic. Bar-Asher Siegal (2013) reports a regular use of the suffixed form *napša-x* as an anaphor referring to humans, alongside what he refers to

as a rare use as an adnominal intensifier, which does not exceed two dozen instances (Bar-Asher Siegal, PC).

This difference is unexpected if the intensifier use predated the anaphoric use, or even if both emerged together. To verify this, in Bassel (in preparation) I compared the anaphor-intensifier ratio found in the Qur'an with two corpora that include both grammatical conventions: EnTenTen21 corpus of Modern English, and the historical corpus of the Hebrew Mishna. The latter is a Rabbinic Hebrew text from the second century CE, which features the earliest uses of Hebrew anaphoric and intensifier *acm-x* (Segal 1935).

For EnTenTen21, I collected all pronoun-*self* expressions and sorted them into post-nominal, post-verbal, and post-prepositional. I classified post-nominal instances as intensifiers and post-prepositional and post-verbal as anaphors, knowing that the latter also includes adverbial intensifiers. The results were 0.452/1000w for anaphors and 0.088/1000w for intensifiers, a significantly lower gap compared to that found in the Qur'an ($\chi^2(1, N = 2650) = 61.213, p < .001$). For the Hebrew Mishna, I classified all instances of *acm-x* according to their meaning in context and found a balanced proportion of 29 anaphors and 32 intensifiers. This comparison suggests that the low proportion of intensive instances of Qur'anic *naf-x* indicates that this meaning was not broadly available when the anaphoric convention emerged.

Despite showing a reversed directionality compared to English, these findings point to another economically motivated process, which resonates with well-known paths of reduction. The pronominal component in the anaphor is reduced to agreement marking. Reduction proceeds when the agreement suffix is completely absent, as seen in pre-nominal *naf-s* modifiers. The full path of reduction for the pronominal component is given in (17).

(17) Pronoun > agreement > $\emptyset$

This proposal does not automatically extend to other Semitic languages, since there is also a possibility for intensifiers to develop directly from body terms through mechanisms of semantic widening. I focused on Arabic since it clearly shows that intensifier uses of suffixed *naf-s-x* emerge in an advanced stage of the anaphoric use, and this might be true for cognates of *naf-s-x* in a broader sense.

3.2.2 Germanic and Romance languages

Romance anaphors typically pattern with English in combining pronominal elements with existing intensifiers (e.g., French *même*, Spanish *mismo*, or Italian *stesso*), reflecting the familiar path from intensifier to anaphor. However, there is some evidence that these

intensifiers have an anaphor in their background, which mirrors the direction of change seen in Arabic.

Romance intensifiers trace back to the Latin intensifier *ipse* (König and Siemund 1999), which originally showed inflection for case, number, and gender, as seen in the attested forms in Table 3.

	Singular			Plural		
	Ms.	Fm.	Neut.	Ms.	Fm.	Neut.
Nom.	<i>ipse</i>	<i>eapse</i>	<i>ipsum</i>	<i>ipsī</i>	<i>eaepsae</i>	<i>ipsa</i>
Acc.	<i>ipsum</i>	<i>eampse</i>	<i>ipsum</i>	<i>ipsōs</i>	<i>ipsās</i>	<i>ipsa</i>
Dat.	<i>ipsī</i>	<i>ipsī</i>	<i>ipsī</i>	<i>ipsīs</i>	<i>ipsīs</i>	<i>ipsīs</i>
Abl.	<i>eōpse</i>	<i>eāpse</i>	<i>ipsō</i>	<i>ipsīs</i>	<i>ipsīs</i>	<i>ipsīs</i>
Gen.	<i>ipsiūs</i>	<i>ipsiūs</i>	<i>ipsiūs</i>	<i>ipsōrum</i>	<i>ipsārum</i>	<i>ipsōrum</i>

Table 3: Declinations of *ipse* in Plautus (Old Latin, Weiss 2009: p.346)

According to Weiss (2009), the inflection paradigm of *ipse* in Old Latin suggests that it is a fusion between an anaphoric demonstrative pronoun *is* and a particle *pse*. Toro et al. (2022) describe pronominal *ipse* as a long-distance anaphor which shares its sources with the Old Latin anaphor *se*. Some authors argue further that *-pse* itself is a compound containing a *se* pronoun (Berenguer 2000; De Vann 2008; Weiss 2009).

Regardless of the exact meaning of early *ipse*, it is a complex pronoun in its morphological analysis, which suggests that all Romance intensifiers have a complex pronoun in their history. The path from *ipse* to modern intensifiers differs between languages and seems to include further iterations of compounding with pronominal elements (e.g., *iste*, *memet*) and emphasizing expressions (Harris 1974; Carlier and De Mulder 2010; Hertenberg 2011; Lüdtke 2015).

There are similar hypotheses with respect to the history of the Germanic intensifier *self*, which is assumed in the reconstruction literature to share a Proto-Indo-European source with the Latin forms *ipse* and *se* (Brugmann and Delbrück 1893; Skeat 1893; Mezger 1948; Hahn 1963; Szemerényi 1964; Erhart 1970; Shields 1998; Petit 1999; Orel 2003). This analysis remains speculative, but English also shows an anaphor-to-intensifier directionality in more recent stages.

Starting 1250 CE, the *x-self* compound has extended from the anaphoric use to the intensifier, at the expense of the existing monomorphemic *self* (Farr 1905). This shift is unmotivated from a morpho-semantic perspective, since the intensifier bares no reference in

either version, which makes the pronominal component semantically vacuous. It is nonetheless predictable given the data from Semitic languages, since it realizes the same meaning extension seen for *nafs-x*. In my understanding, the anaphoric form extends to the intensifier because anaphors have the semantic capacity to generate intensifier meanings.

There is still a question of why, given two possible forms of intensifiers, the complex form was preferred over the more economic monomorphemic version. There seems to be an arbitrary component to this process: In English, the complex intensifier fully replaced the simple variant, but other Germanic languages retained their monomorphemic intensifiers, such as German *selbst*, Danish *selv*, or Norwegian *själv*. This variability is unexpected in an economical account, which predicts that a more simple form would be preferred categorically out of two semantic equivalents.

The fact that this shift happened for English nonetheless suggests that the complex intensifier is not entirely equivalent to the simple variant. More specifically, it suggests that speakers were motivated to renew the agreement marking of the intensifier, which was reduced in the course of the compounding process, retaining only the number distinction (*self~selves*). A richer agreement could have played a role in facilitating the connection between the intensifier and the head noun, as recently demonstrated for verb-subject agreement (Keshev and Meltzer-Asscher 2024). This is particularly relevant for adverbial intensifiers, which appear at a distance from their antecedent, as illustrated by the following Modern English example.

(18) The queen signed the letters **herself**.

Evidence for a similar transition to complex intensifiers are found in French. A search of Frantext corpus of French literature shows that, in Middle French, the form *mesme* occurs as an intensifier and as modifiers meaning ‘even’ and ‘same’. By the 19th century, its reduced successor *même* is still used as ‘even’ and ‘same’, alongside a regular use as part of the complex anaphor *x-même*. At this stage, the monomorphemic *même* is the common choice for an intensifier, but there are also intensive instances of the complex variant, mainly for adverbial intensifiers. The following quotes show both forms used within the same sentence, the simple *même* as an adnominal intensifier and the complex *lui-même* as an adverbial one.

(19) **Soliman même** n'a tant d'éclat, que parce que le poëte lui a donné quelques traits de la générosité du chevalier : ainsi le principal héros infidèle **emprunte lui-même** sa majesté du christianisme

‘Suleiman himself has so much brilliance only because the poet has given him some

traits of the knight's generosity: thus the principal infidel hero borrows himself his majesty from Christianity.'

(François-René de Chateaubriand, 1828, "Génie du christianisme", p.192)

In contemporary French, *même* and *x-même* are both in use as adnominal intensifiers, as illustrated in the following attested examples. French speakers I consulted with reported that the monomorphemic variant sounds more formal.

(20) Ne pas confondre imposte et oculus. Ce dernier est une ouverture vitrée aux formes variées, pratiquée dans **la porte même**.

'Do not confuse impose and oculus. The latter is a glass opening featured in the door itself.'

(www.m-habitat.fr/portes/elements-d-une-porte/l-imposte-d-une-porte-2786_A)

(21) La certification A2P prend en compte **la porte elle-même** mais aussi la serrure.

'A2P certification takes into account the door itself but also the lock.'

(www.futura-sciences.com/maison/questions-reponses/bricolage-porte-blindee-choisir-4254)

Considering these facts together, the emergence of anaphors, intensifiers, and 'same' particles in French is in line with the progression suggested for Arabic in the previous subsection. The crucial difference is that French inherits an intensifier bundled with the adjectival 'same' meaning from Latin, while Arabic starts the process from a lexical source and acquires the intensive and 'same' meanings at later stages. Table 4 summarizes.

Anaphor		Intensifier		'same'
Qur'anic Arabic: <i>nafs-x</i>	>	Classical Arabic: <i>nafs-x</i>	>	Modern Standard Arabic: <i>nafs</i>
		Middle French: <i>mesme</i>	~	Middle French: <i>mesme</i>
Early Modern French: <i>x-même</i>	>	Modern French: <i>x-même</i>		

Table 4: Anaphors, intensifier and 'same' in the history of Arabic and French

To conclude, the path from intensifiers to anaphors is well-documented, and yet there is also evidence for a path from anaphors to intensifiers, which proceeds to focus and identity particles.

3.3 Anaphors to middle markers

The previous subsections introduced shifts between intensifiers and anaphors in both directions. The current one will center on the common link between anaphors and markers of middle voice (Faltz 1977, Kemmer 1993) and highlight intersections between these processes.

The hypothesis that anaphors evolve into middle markers is based on observations that languages unrelated as German (Germanic), Pangwa (Bantu), and Guugu Yimidhirr (Australian) show identity between these morphemes. In other languages, such as Russian (Slavic) and Surselvan (Romance), the middle marker is a reduced variant of the anaphor. A transition from anaphors into middle markers is economic at the level of the VP, since it turns a verb phrase into a verb head. Certain languages have lost the overt middle marker, like Modern English, which marks the end of the cycle. This progression echoes a general course of reduction, seen in (22).

(22) *Anaphor to Middle Voice:*

Pronoun > clitic > affix > $\emptyset$

On the semantic level, the shift from anaphor to middle raises the level of abstraction from coreference to indistinguishability of event participants. For a verb with two prototypical roles – a causer and an affectee – an anaphor assigns both roles to the same entity, while middle voice contracts them into one role that does not clearly satisfy either prototype. This difference is reflected in the anaphor's independent status as an argument, which allows it to generate the so called proxy readings which refer to statues or pictures, among other diagnostics (Ruigendijk and Schumacher 2020; Maddox 2021; Bassel 2024). The fact that middle markers lack these capacities explains why numerous languages follow the transition from anaphors to middles with a renewal of the anaphoric form.

Romance languages present a case in point. Latin featured two distinct constructions: an anaphor *sē* showing characteristics of an independent DP, and a middle marker *-r* used in a variety of change-of-state events featuring one entity (Kemmer 1993). In the shift to Romance, successors of *sē* gradually replaced the middle marker across its various uses (Hatcher 1942), and new anaphors emerged from combinations of pronouns and intensifiers (Kemmer 1993; Waltereit 2012; Maddox 2021). The following example from Spanish demonstrates a co-occurrence of a *se* middle marker and a renewed anaphor *si mismo* (23), which indicates that the older anaphor completed the cycle (at least in this context).

(23) *Middle Spanish:*

si es necçessario que el onbre se ame a si mismo mas que
 if is necessary that the man MIDDLE loves ACC REFL very more than
 a los otros onbres.

ACC the other men

‘...if it is necessary that one love himself more than others.’ (Maddox 2021: 47)

Kemmer presents a similar cycle in Nilo-Saharan languages (Crazzolara 1955; Kemmer 1993). In particular, cognates of the body-part morpheme *rô* ‘body’ are used across Nilo-Saharan as anaphors and intensifiers, with reduced variants surfacing as verbal affixes. One of these suffixes is found in Àcoólî, where *rô* has been reduced into a single vowel -ε. Àcoólî also includes a renewed compound based on the morpheme *kööm* ‘body’, used as an anaphor and an intensifier. The following sentences show the middle marker and the anaphor side by side (24a), and an intensifier use of the same form (24b).

(24) (a) òjùkù **kööm**-ε = òjùk-ε

paint body-3SG.M paint-MIDDLE

‘he painted himself (with mud)’

(Kemmer 1993: 112)

(b) aan ki **kööm**-a

I with body-1SG

‘I myself’

These examples show that reduction of the anaphoric capacity may lead to regeneration based on intensifiers or body terms.

3.4 The cycle

The processes outlined above demonstrate three economically motivated shifts, listed below: compounding of a pronoun and an intensifier that produces an anaphor (25), reduction of anaphors to intensifiers (26), and another course of reduction to middle marker (27).

(25) *Intensifier to anaphor:*

[_{DP} [_D pronoun] [_{AP} intensifier]] > [_{DP} anaphor]

(26) *Anaphor to intensifier:*

Pronoun > agreement > ∅

(27) *Anaphor to Middle Voice*:

Pronoun > clitic > affix > $\emptyset$

In the sources I based on, the anaphors that interact with intensifiers (25-26) are morphologically different from the ones that reduce to middle markers (27). The former clearly present compounding of a pronoun with an additional morpheme, while the latter are generally considered monomorphemic. Treating these classes as part of the superset “anaphors” allows to combine the three processes into the cycle shown in Figure 4.

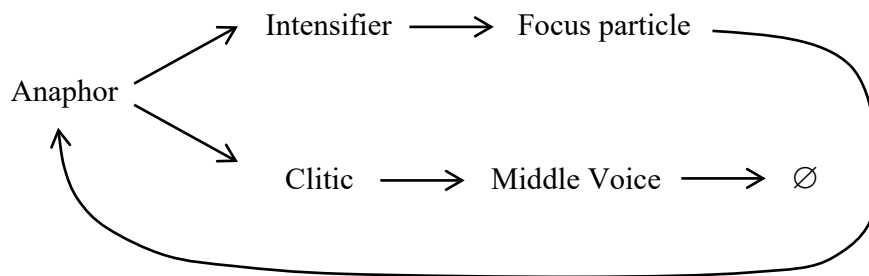


Figure 4: A reflexive-intensive cycle (extended)

Since every step of this process is economically motivated, it is expected to be unidirectional. The cyclic nature follows from the tendency of languages to regenerate an anaphoric element with particular properties. The history of Romance anaphors showcases all stages of the proposed cycle, excluding the edges (lexical source and full reduction). Figure 6 demonstrates this for French.

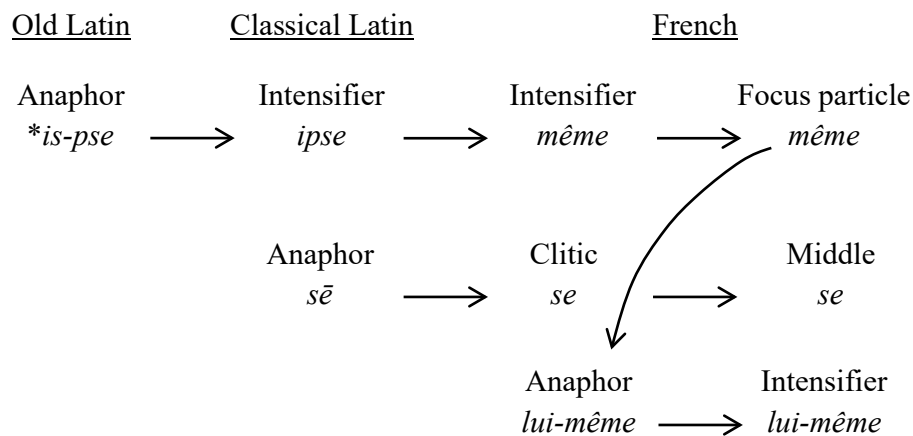


Figure 6: Cyclicity in Latin and French anaphors

The proposed cycle explains the recurrence of this network of elements across unrelated languages, and the differences between anaphors that emerge from a lexical

source (Faltz's "head reflexives") and ones that are renewed based on an existing intensifier.

4. Conclusion

I described documented processes of language change with independent motivations reported in the literature: intensifiers turn into anaphors to reduce ambiguity in the third person, anaphors turn into intensifiers due to shared semantic properties or reduce to middle markers for reasons of economy. Two of these processes go in opposite directions (intensifiers to anaphors, anaphors to intensifier) and could have been interpreted as a reversal of the historical process. I propose instead to consider these procedures as parts of a cycle, in which each step is explained by economy principles.

The proposed account solved a puzzle that has been unapproachable for syntactic theory, namely why are there many expressions that share form and function with anaphors and recur in unrelated languages. From a synchronic standpoint, the existence of such elements is a burden that obscures the regularity of pronominal systems. A reflexive-intensive cycle of language change predicts that these elements should exist and explains the points of similarity and variation between them.

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